

Which Numerical Differentiation Method Should We Trust?

Investigate the accuracy of three finite difference methods for numerical differentiation. Rather than simply implementing the formulas, you will compare their performance for different functions and different step sizes, analyze the errors, and identify the most reliable method.

Implement the following numerical differentiation methods:

- 1 Forward Difference
- 2 Backward Difference
- 3 Central Difference

Compare their performance using three test functions.

- $f(x) = e^x$
- $f(x) = \sin x$
- $f(x) = x^3 - 2x + 1$
- Use at least six different values of h (e.g., 10^{-1} , 10^{-2} , ..., 10^{-6})
- Evaluation Point: Approximate $f'(1)$ for all three functions.
- Compute: $Absolute\ Error = |f'_{exact} - f'_{numerical}|$

Prepare:

- a results table,
- a log-log error plot,
- and answer the following:
 - Which method is consistently the most accurate?
 - Does the observed error decrease at the expected rate?
 - Is there a value of h beyond which the error stops decreasing?

Log-log error plot: y-axis = $\log(\text{error})$, x-axis = $\log(h)$

Additional question: Can you derive the forward, backward & central difference formulas symbolically for this polynomial & identify the error term?

$$f(x) = x^3 - 2x + 1$$